Vishal V Devan

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Summary

Full-Stack Systems Engineer and 2nd year CSE honors student with a track record of architecting scalable, edge-native web infrastructure. Specialist in high-performance web runtimes and distributed systems, having engineered real-time platforms using Hono.js, Cloudflare Workers, and PostgreSQL. National-level hackathon winner, with proven expertise in building production-ready backends and optimizing serverless workflows. Passionate about applying systems-level performance and memory safety to modern full-stack development.

Education

Muthoot Institute of Technology and Science, Kochi — Expected 2028
B.Tech in Computer Science and Engineering — *Kerala, India*

- Academic focus on Computer Architecture, Theory of Computation, Advanced Data Structures
- Enrolled in Honors program, current CGPA 9.3

Experience

Full-Stack Architect / Systems Engineer (Independent) — Oct 2020 – Present
Remote

- Architected scalable web systems and real-time platforms using **Hono.js**, **Cloudflare Workers**, and **PostgreSQL** for high-concurrency environments.
- Engineered high-performance backend infrastructure with a focus on edge-native distribution, type-safe schemas, and bridging systems-level optimization with modern web runtimes.

Achievements

Second Place — AI Samasya National Hackathon — 2025
Undergraduate • 75 Teams

- Placed **2nd** nationally out of 75 teams by building a regression and generative-AI system for early detection of learning disabilities.
- Designed data pipelines, evaluation strategy, and interpretable models under strict time constraints.

First Place — PALS Ideathon, IIT Madras (Inter-College) — 2024
Technical Problem-Solving Challenge

- Won first place by decomposing an open-ended engineering problem into solvable components and iteratively optimizing solutions.
- Performed feasibility analysis and quantitative trade-off evaluation in a 48-hour on-site competition.

Technical Skills

Languages: TypeScript, JavaScript, C, Java, Python
Systems & Concepts: Computer Architecture (RISC-V), Programming Language Design, Recursive Descent Parsing, Interpreter Implementation, Web Performance Optimization, Distributed Web Systems
Frameworks & Tools: Next.js, NestJS, Hono, PostgreSQL, Git, Docker, Linux

Projects

Real-Time Streamer Tipping and Live Chat Platform — 2026

- Engineered a low-latency chat and donation system using **Cloudflare Workers** and **WebSockets** to support real-time bidirectional communication.
- Implemented **Durable Objects** to maintain globally consistent state for live sessions and coordinated high-concurrency interaction events.
- Designed an automated **real-time moderation queue** and optimized data retrieval using **Workers KV** for high-performance caching at the edge.

Medleaf PatientManager — EHR System — 2024

- Developed a production-grade EHR web app PWA using Next.js, PostgreSQL, and Drizzle ORM.

- Deployed in several real world clinics and optimized for high load on servers in serverless settings
- Implemented role-based access control, offline-safe caching, and responsive UX.

AWL — Custom Interpreted Programming Language 2025

- Implemented a custom interpreted language from scratch with a hand-written recursive descent parser and AST-based execution.
- Designed scoped environments, control flow, and runtime semantics for real-world automation use cases.

Dominity — Lightweight Frontend Framework 2023

JavaScript

- Built a **sub-10 KB** signal-based reactive frontend framework with aggressive runtime and bundle-size
- Achieved near–industry-standard performance, implemented a minimal router, and shipped as an **npm package**.

Extracurricular Activities

Tech Lead — TinkerHub, kMITS

- Served as Tech Lead for the TinkerHub MITS, contributing to technical direction and peer learning initiatives.
- Mentored participants across multiple hackathons, guiding teams through architecture decisions, debugging, and deployment workflows.
- Helped organize and coordinate **hackathons and technical events**, supporting problem statements, mentoring, and execution.
- Conducted **hands-on sessions** for juniors on Linux, Docker, and modern development workflows, focusing on practical tooling and best practices.
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Vice President Membership — Toastmasters of MITS

- Led membership growth and retention initiatives, managing onboarding pipelines and engagement strategies.
- Coordinated guest outreach, follow-ups, and induction processes to ensure smooth member transitions into active roles.
- Collaborated with executive committee to plan meetings, track member progress, and strengthen overall club participation.